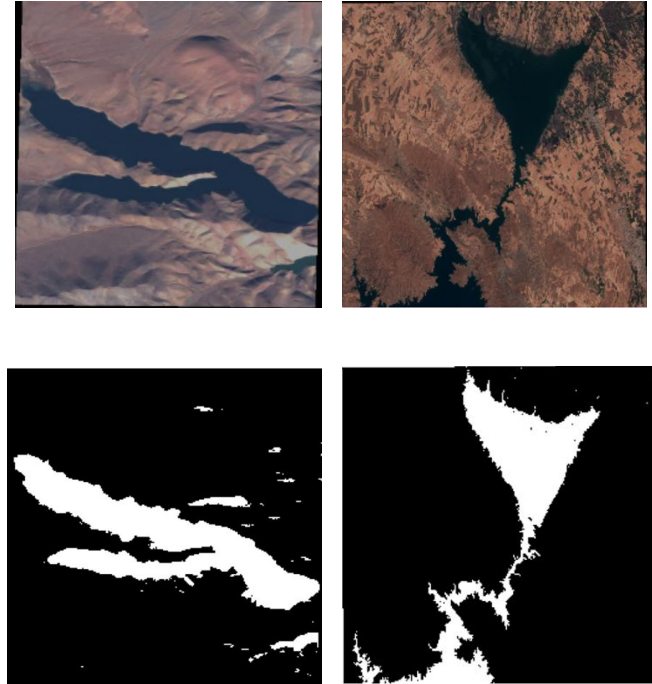




Neural Networks for Water Body Segmentation

Ludovico De Salvo
Clara Gonçalves

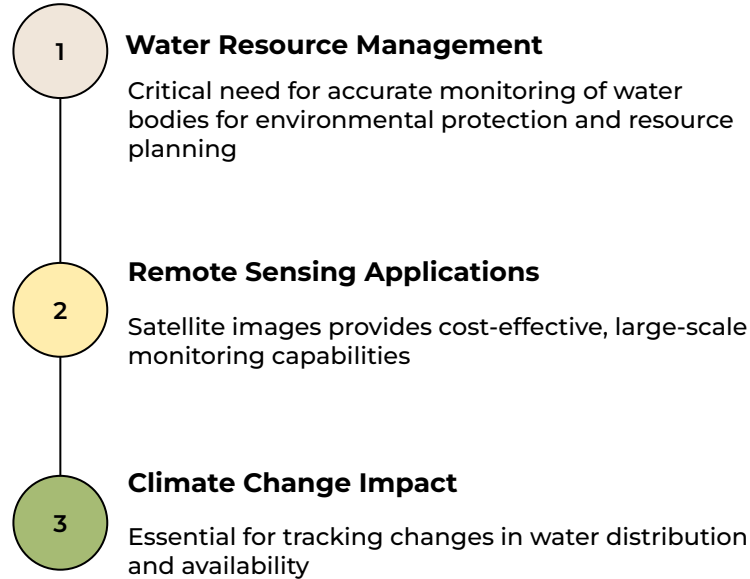
Neural Networks
26 June 2025



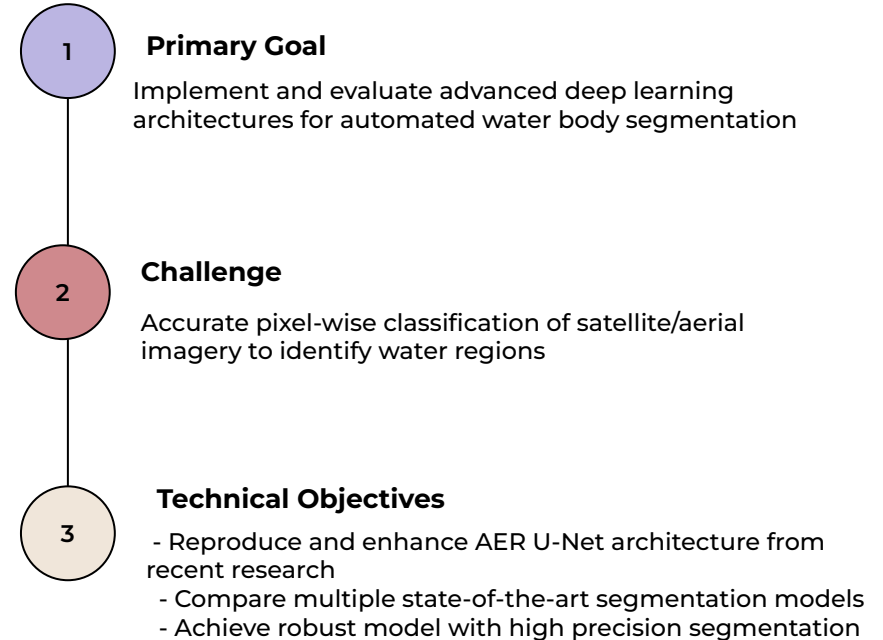
Based on the paper: [AER U-Net: Attention-Enhanced Multi-Scale Residual U-Net Structure for Water Body Segmentation Using Sentinel-2 Satellite Images](#)(Nature Scientific Reports, 2024)

Motivation and Aim

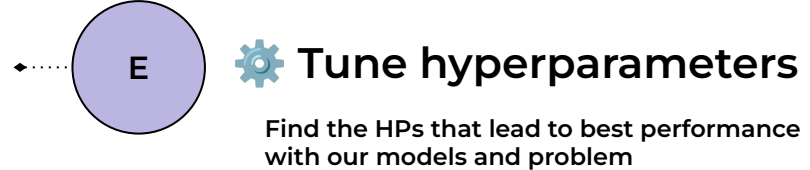
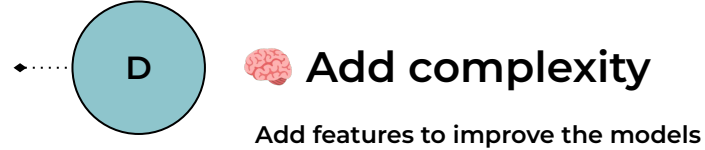
Motivation



Aim



Method overview



Implementation and experiments

Core Architectures

U-Net

Encoder-decoder with skip connections

U-Net++

Nested dense skip pathways with deep supervision

AER U-Net

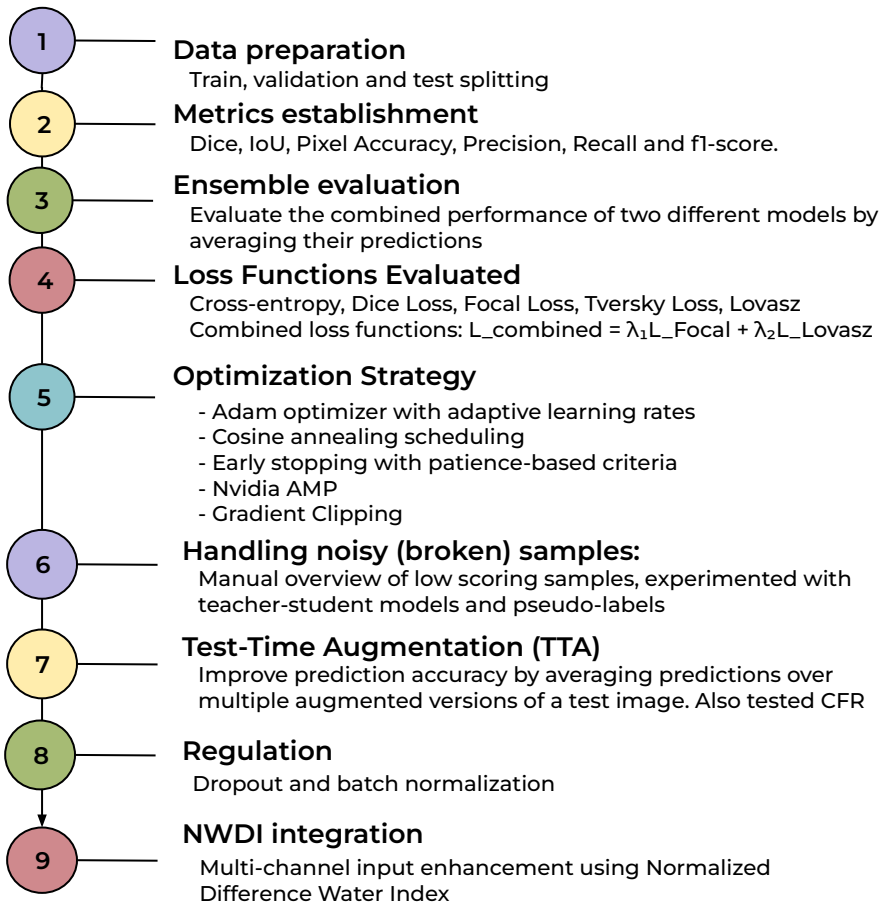
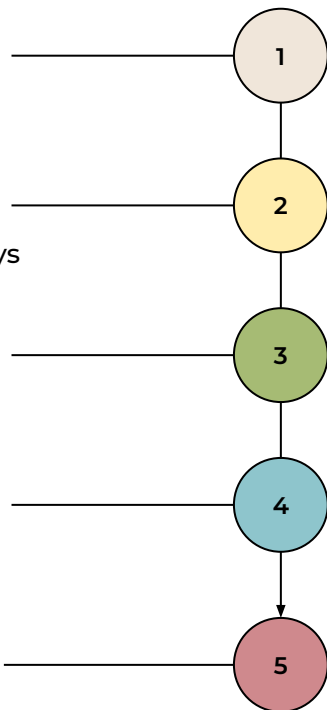
Attention-enhanced with residual blocks

SegFormer

Transformer-based hierarchical approach

CLI Wrapper

Runtime and models, training, evaluation, data preparation all handled dynamically with logging



Results

Dice Score:	Mean: 0.9343 ± 0.0635 Range: [0.6608, 1.0000]
Iou Score:	Mean: 0.8827 ± 0.1017 Range: [0.4934, 1.0000]
Pixel acc Score:	Mean: 0.9677 ± 0.0317 Range: [0.7965, 1.0000]
Overall Dataset Metrics:	Precision: 0.9587 Recall (Sensitivity): 0.9405 F1-Score: 0.9495

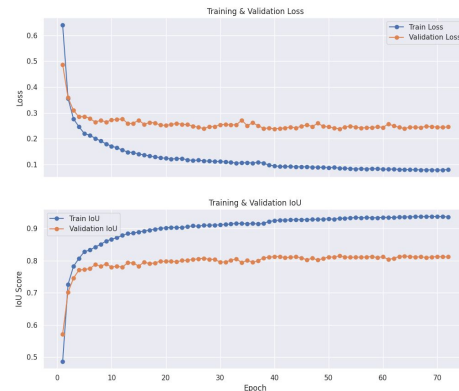
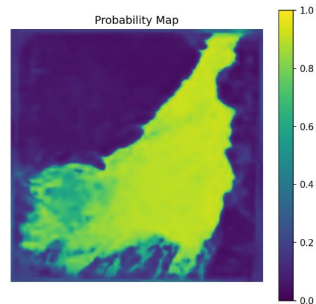
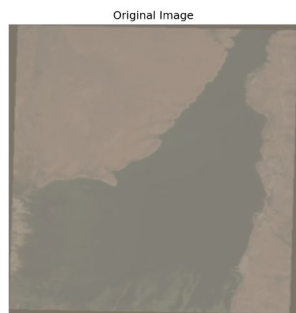
Comparison IoU Score:

Baseline Unet: 0.7192 ± 0.2086

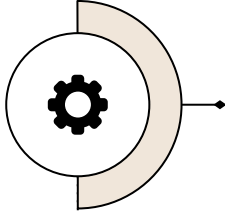
Unet++ with nwdi: 0.8169 ± 0.1483

Ensemble unet++ and segformer (no nwdi): 0.8153 ± 0.1850

With noisy mask cleaning (no nwdi): 0.8361 ± 0.1499

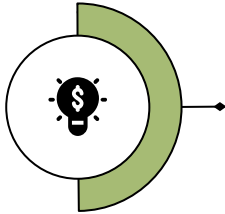


Conclusion



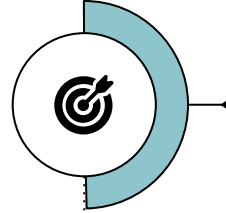
Goal Achievement

We successfully developed a robust and flexible end-to-end pipeline for water body segmentation. The project is highly usable thanks to its interactive menu and modular structure.



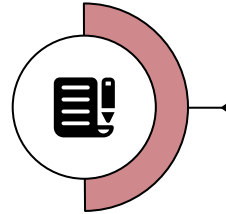
Key Technical Takeaways

- **Attention mechanisms:** significantly improve water boundary detection accuracy
- **Residual connections:** enable training of deeper, more effective networks
- **Advanced loss functions:** (FocalLoss) effectively handle class imbalance
- **Ensemble methods:** provide additional performance gains through model averaging



Performance

The implemented models achieved high average scores, demonstrating their effectiveness for this task. The final **F1-Score** of **0.9322** indicates a highly reliable model with significant robustness.



Impact and applications

Real-time Disaster Management: assessing flood extent after major storms to guide emergency response.

Environmental Science: tracking long-term changes in lakes, rivers, and coastlines, offering valuable data for climate change studies and sustainable water resource management.

Adaptable Geospatial Analysis: Flexible framework can be adapted to other remote sensing challenges.

Limitations and suggestions for the future

